

Day 18 Student Handout

Geometry Summer Credit | Chords, Secants, Tangents, and Segment Relationships

Name:	Date:	Period:
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Learning Targets

I can distinguish chords, secants, and tangents. I can use tangent-radius and equal-tangent relationships. I can select and solve the correct segment-product equation from a diagram.

Lesson 18A: Identify the Configuration

Warm-Up

1.	Solve: $6(8)=4x$.
2.	Solve: $x^2=144$ for a positive length x .
3.	An inscribed angle intercepts a 116-degree arc. Find the angle.

Circle Lines

A chord is a segment with two endpoints on the circle. A secant line crosses the circle at two points. A tangent line touches the circle at exactly one point, called the point of tangency.

Problem	Work / notes
A radius is drawn to a point of tangency. What angle is formed with the tangent?	Relationship: Angle:
From external point P, tangent segments PA and PB touch the same circle. If $PA=17$, find PB.	Theorem: Answer:
From external point P, tangent lengths are $3x+4$ and $5x-10$. Find x and each tangent length.	Equation: $x=$ Length:

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Lesson 18B | Segment Products

Choose the Product Rule

Three Configurations

Inside: chord part times chord part = chord part times chord part. Outside with two secants: external times whole = external times whole. Outside with tangent and secant: $\text{tangent}^2 = \text{external times whole}$. Whole secant = external + internal.

Problem	Work / notes
Two chords intersect inside. One has parts 6 and 8; the other has parts 4 and x. Find x.	Rule: Equation: Answer:
Two secants share an external point. First: external 3, whole 15. Second: external 5, whole x. Find x.	Rule: Equation: Answer:
A tangent has length 12. A secant from the same point has external part 8 and whole length x. Find x.	Rule: Equation: Whole/internal:

Decision Routine

Before Multiplying

1. Is the intersection inside or outside? 2. Count tangents and secants. 3. Label external, internal, and whole lengths. 4. Write the matching rule before substituting. 5. Reject negative length solutions.

4.	Explain why an external-secant formula uses the whole secant rather than only the internal portion.
5.	A secant has external part 4 and internal part 9. What number should be used as the whole length?